

University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

Clustering

Moo K. Chung
University of Wisconsin-Madison
www.stat.wisc.edu/~mchung →
github.com/laplcebeltrami

Learning goal

Understand how clustering works

Understand relationship between clustering and statistical inference

Understand state space modeling

Magnetic resonance imaging (MRI)





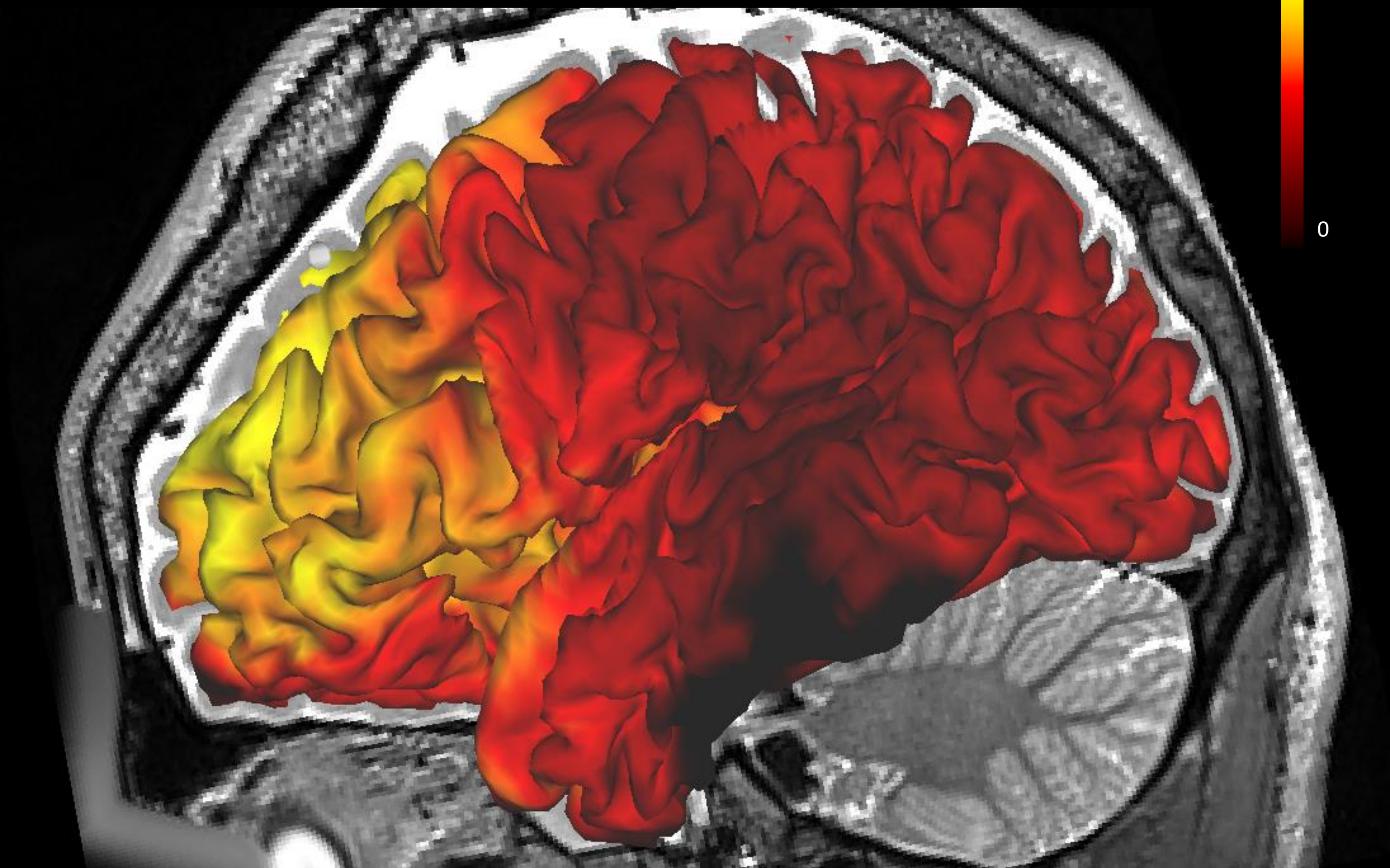
3T GE Discovery X750
Waisman Brain Imaging Laboratory
University of Wisconsin-Madison



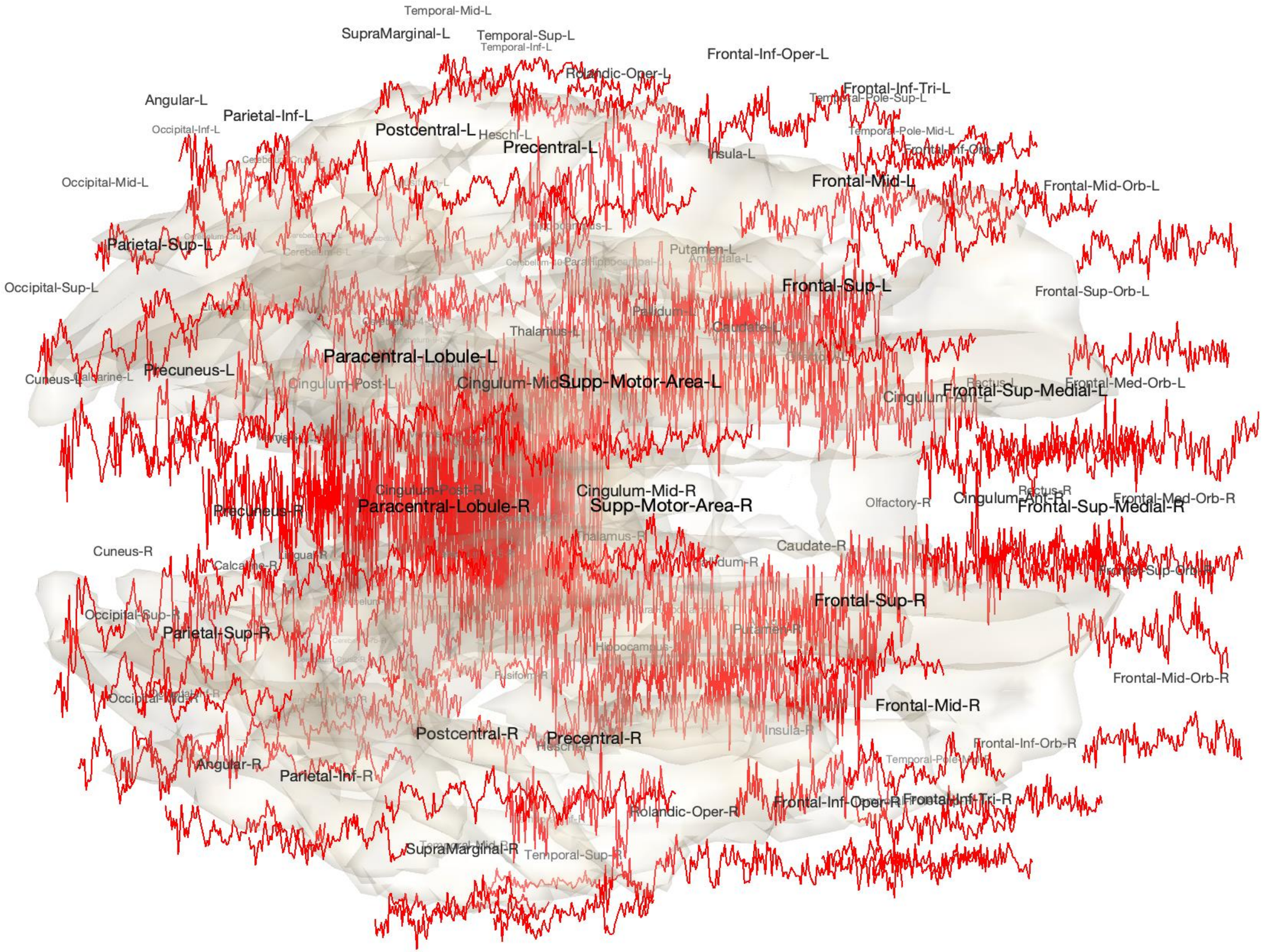


3T GE Discovery MR750
Center for Imaging Research
Medical College of Wisconsin, Milwaukee, WI

rs-fMRI (every 30 second)



Time series averaged into 116 brain r

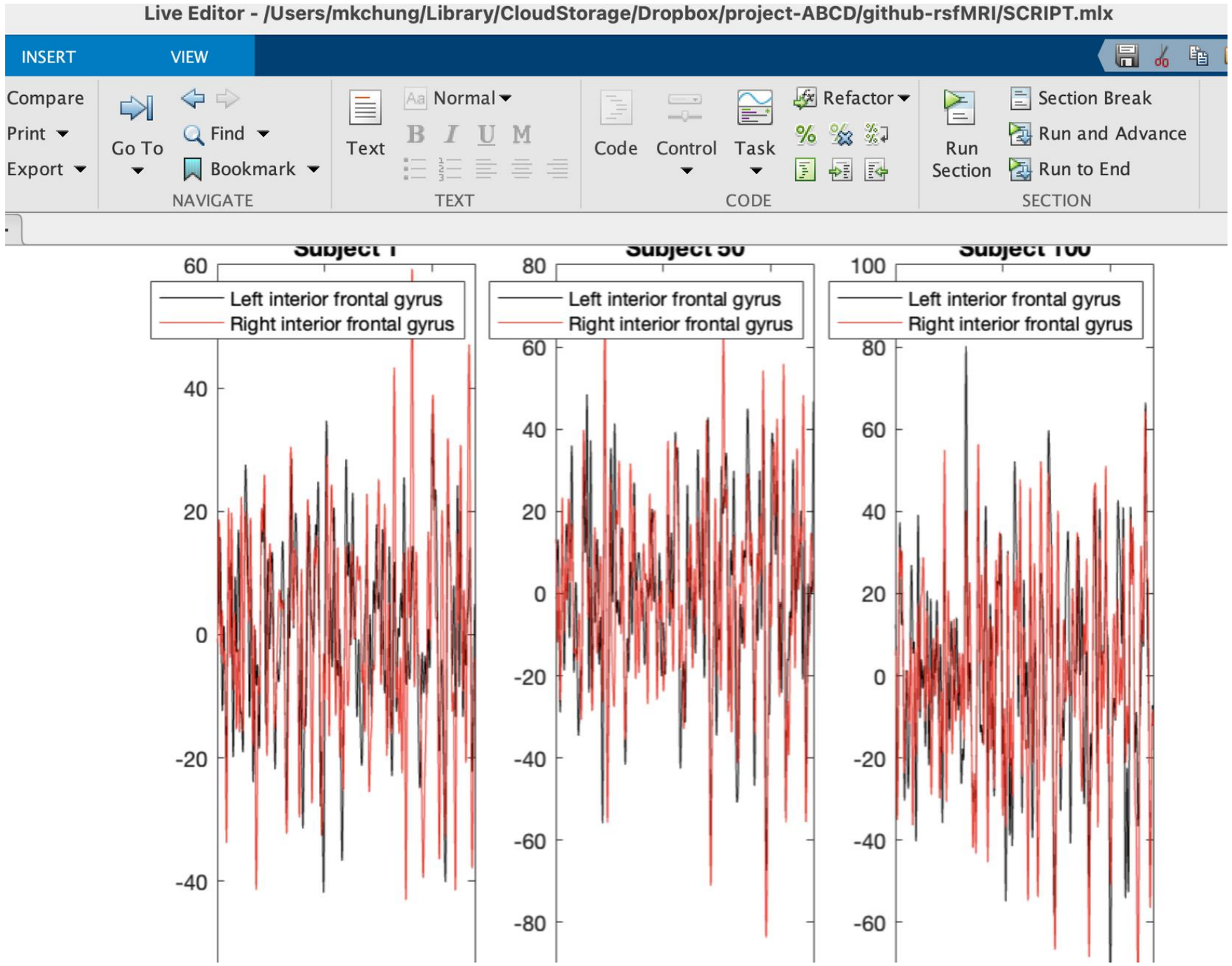


Sample rs-fMRI time series data

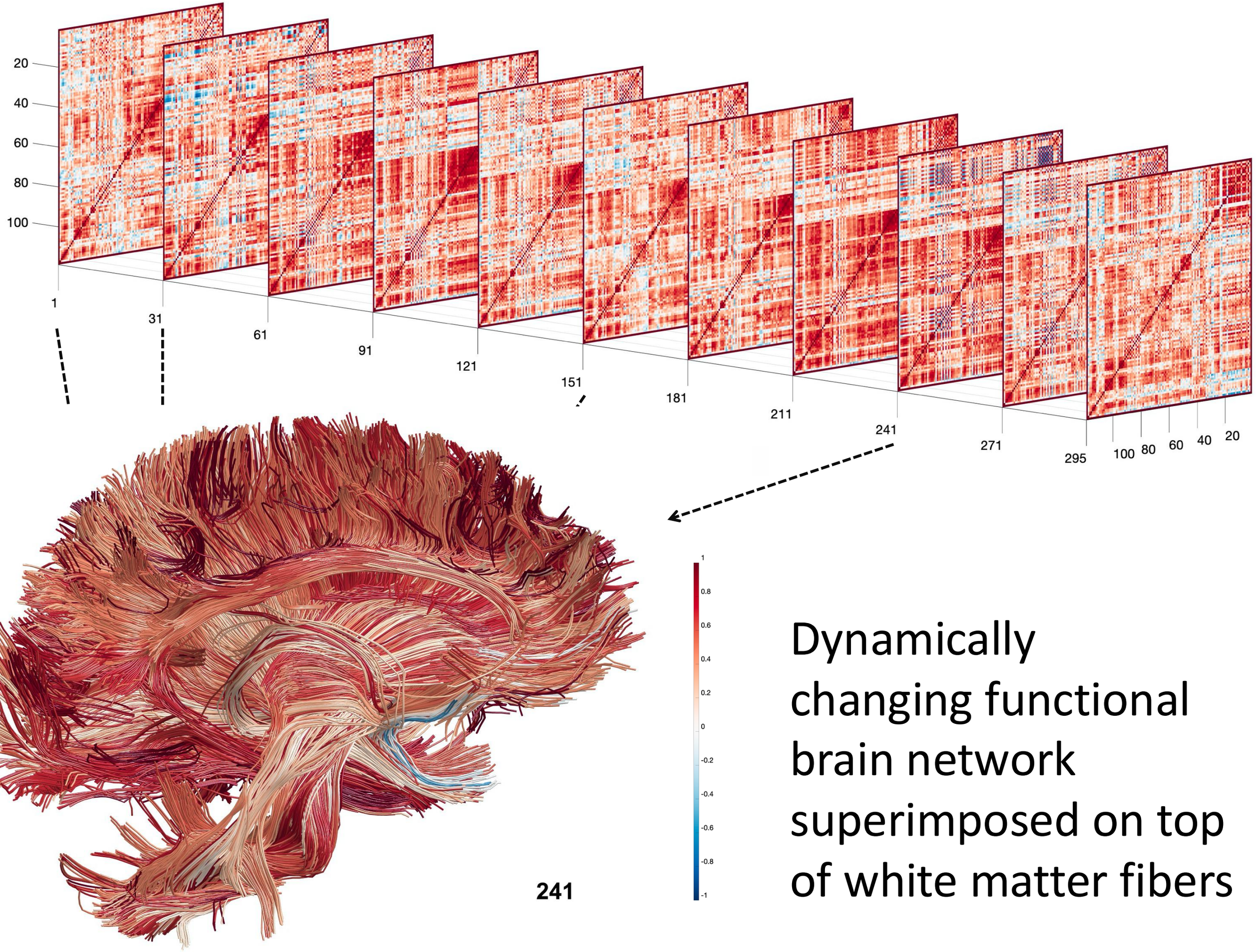
https://github.com/lapalmer/rsfMRI

https://github.com/lapalmer/BM768/tree/main/rsfMRInetwork

Possible project questions are added



Huang et al. 2020 Neuroscience Methods 331:10848



Dynamically changing functional brain network superimposed on top of white matter fibers

Distance between clusters

$$C_1 \cup C_2 \cup \cdots \cup C_k = \{\mathcal{X}_1, \cdots, \mathcal{X}_n\}, \quad C_i \cap C_j = \emptyset$$

Between-group distance

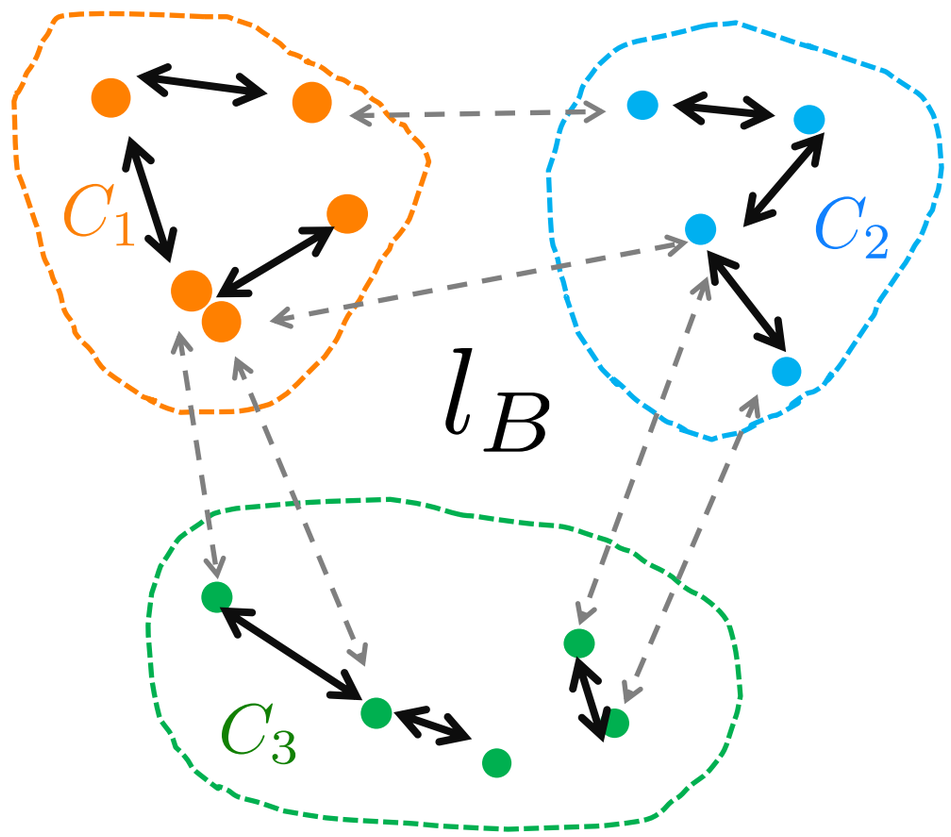
$$l_B \propto \sum_{i \in C_1, j \in C_2} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

$$l_W$$

Within-group distance

$$l_W \propto \sum_{k} \sum_{i,j \in C_k} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

$$l_B + l_W = \sum_{i,j} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$



Clustering

Minimize the within cluster distance over all possible clustering configurations

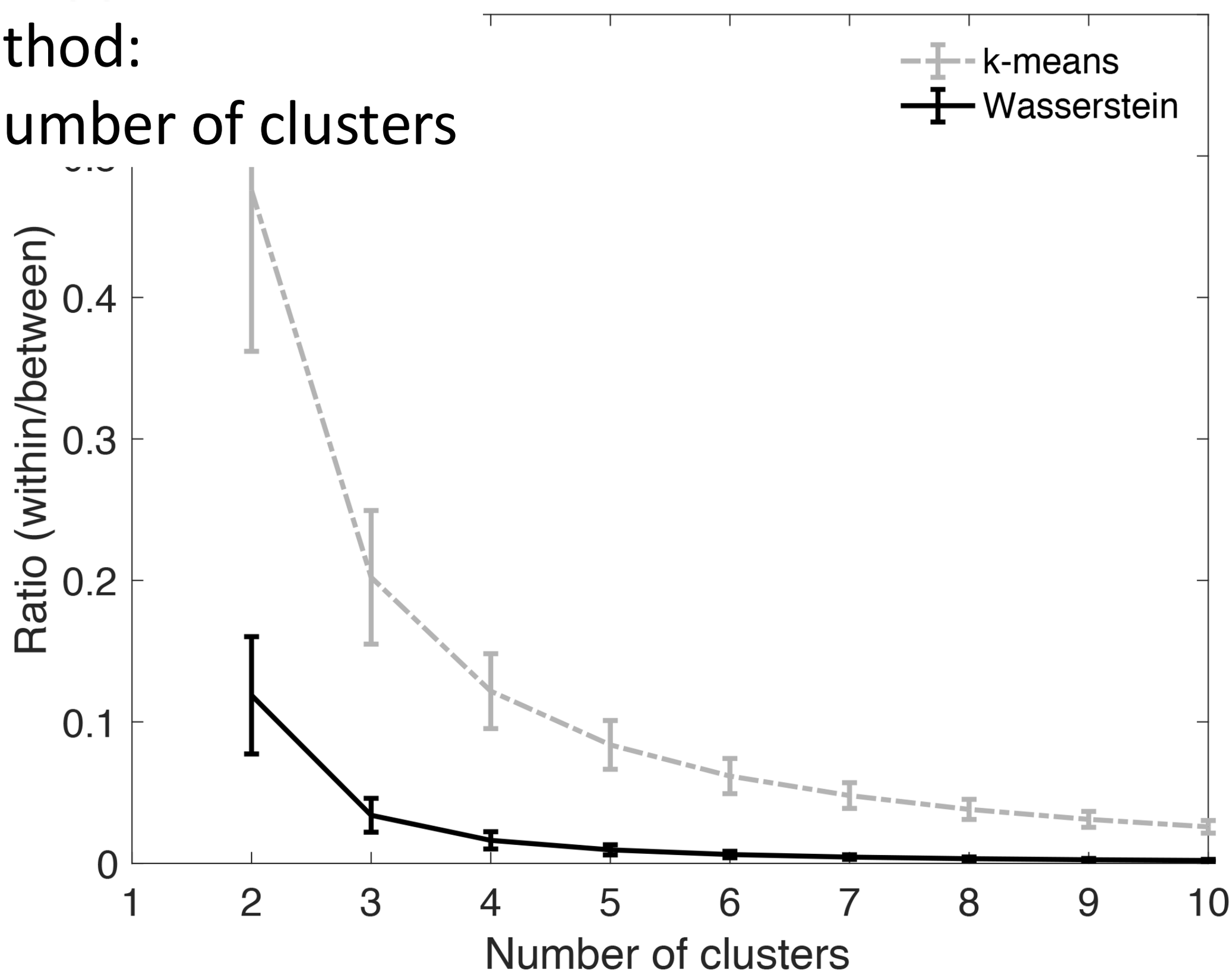
$$l_W \propto \sum_k \sum_{i,j \in C_k} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

Equivalently, maximize the between cluster distance

$$l_B = \sum_{i \in C_1, j \in C_2} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

Elbow method:
Optimal number of clusters

$$\frac{1}{\phi} = \frac{l_W}{l_B}$$



The within cluster variance *6 times* smaller

Self exercise problem

Investigate how clustering accuracy is computed.

Clustering accuracy measures how well the estimated cluster assignments match the true labels. Because clustering is obtained through data-driven estimation rather than deterministic labeling, the reported accuracy reflects **statistical estimation performance** rather than an exact or deterministic quantity.

Mathematical equivalence of topological clustering and topological inference

There exists a monotonically decreasing function f satisfying

p-value = f (clustering accuracy)

Inference

Clustering

Chung et al. 2023 NeuroImage

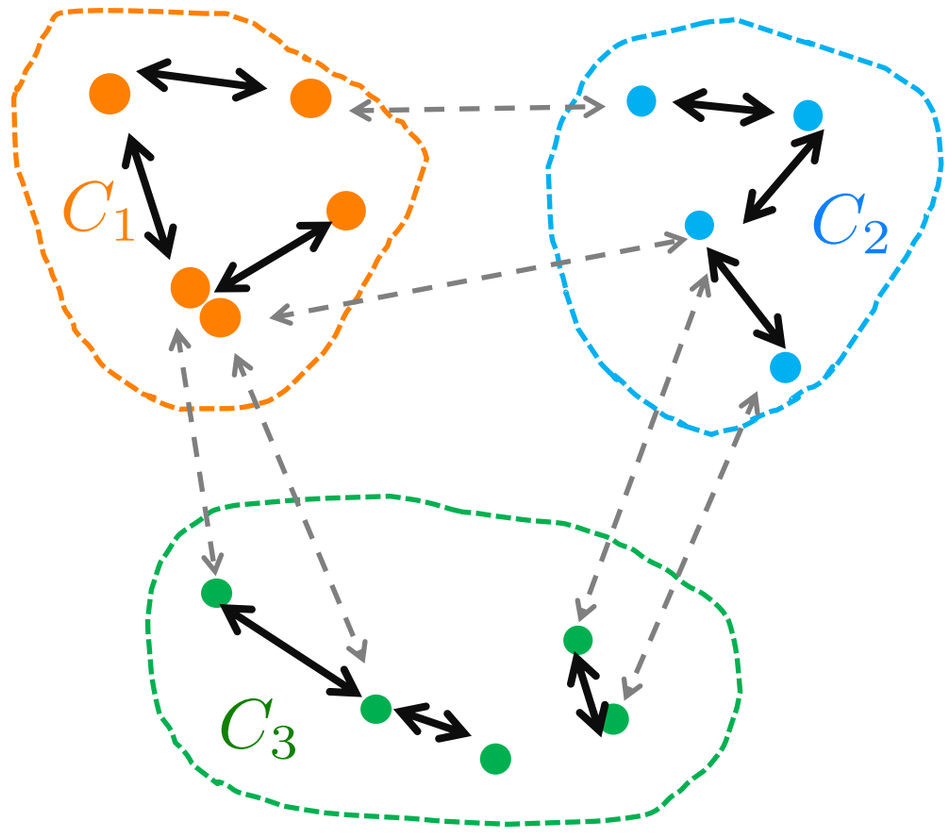
Topological clustering = topological inference

Within-group distance $d_W \propto \sum_k \sum_{i,j \in C_k} d_0(\mathcal{X}_i, \mathcal{X}_j) + d_1(\mathcal{X}_i, \mathcal{X}_j)$

Clustering accuracy

$$\frac{dA}{dp} = \frac{dA}{d(dl_W)} \frac{d(dl_W)}{dp} \leq 0$$

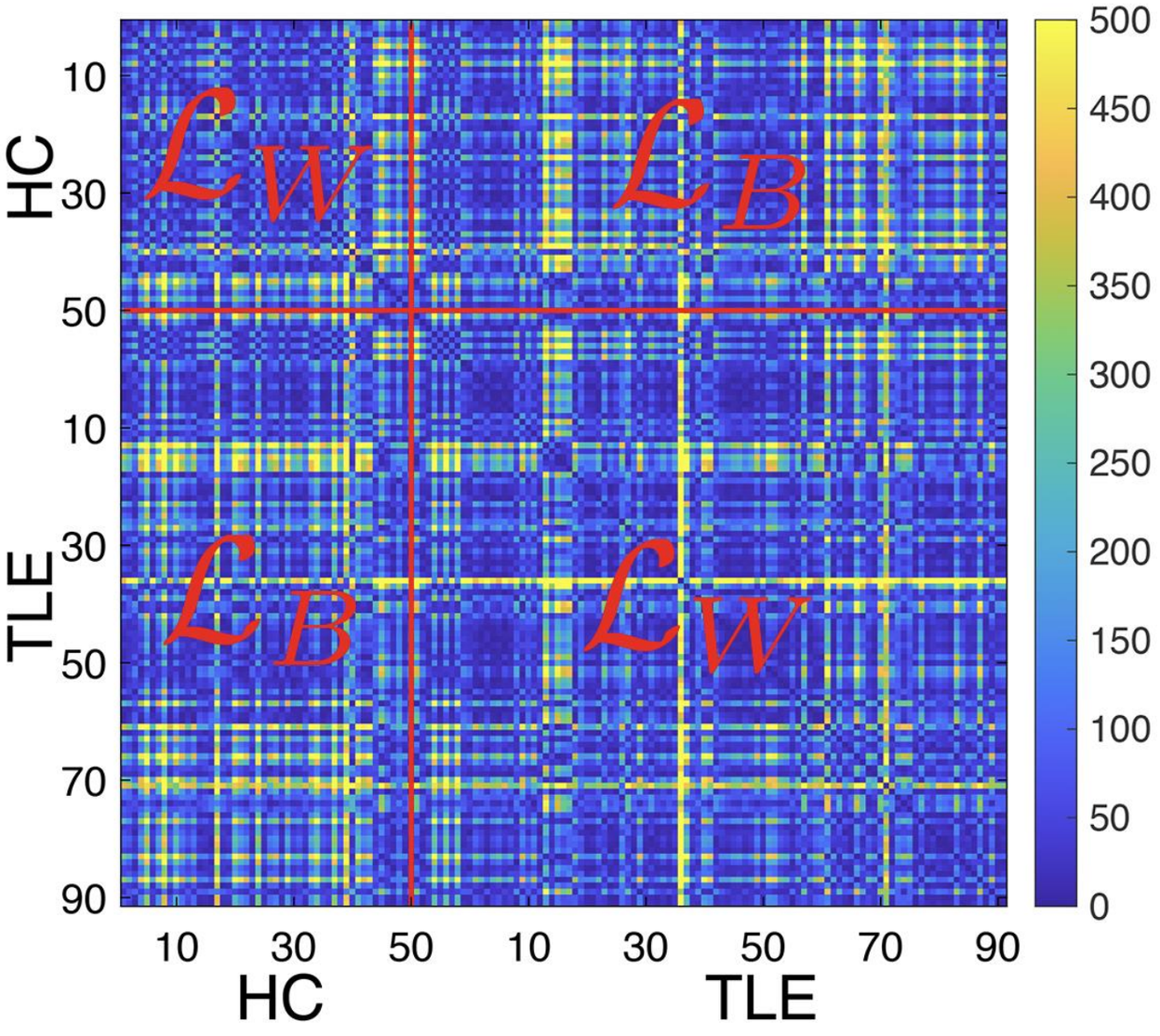
p-value

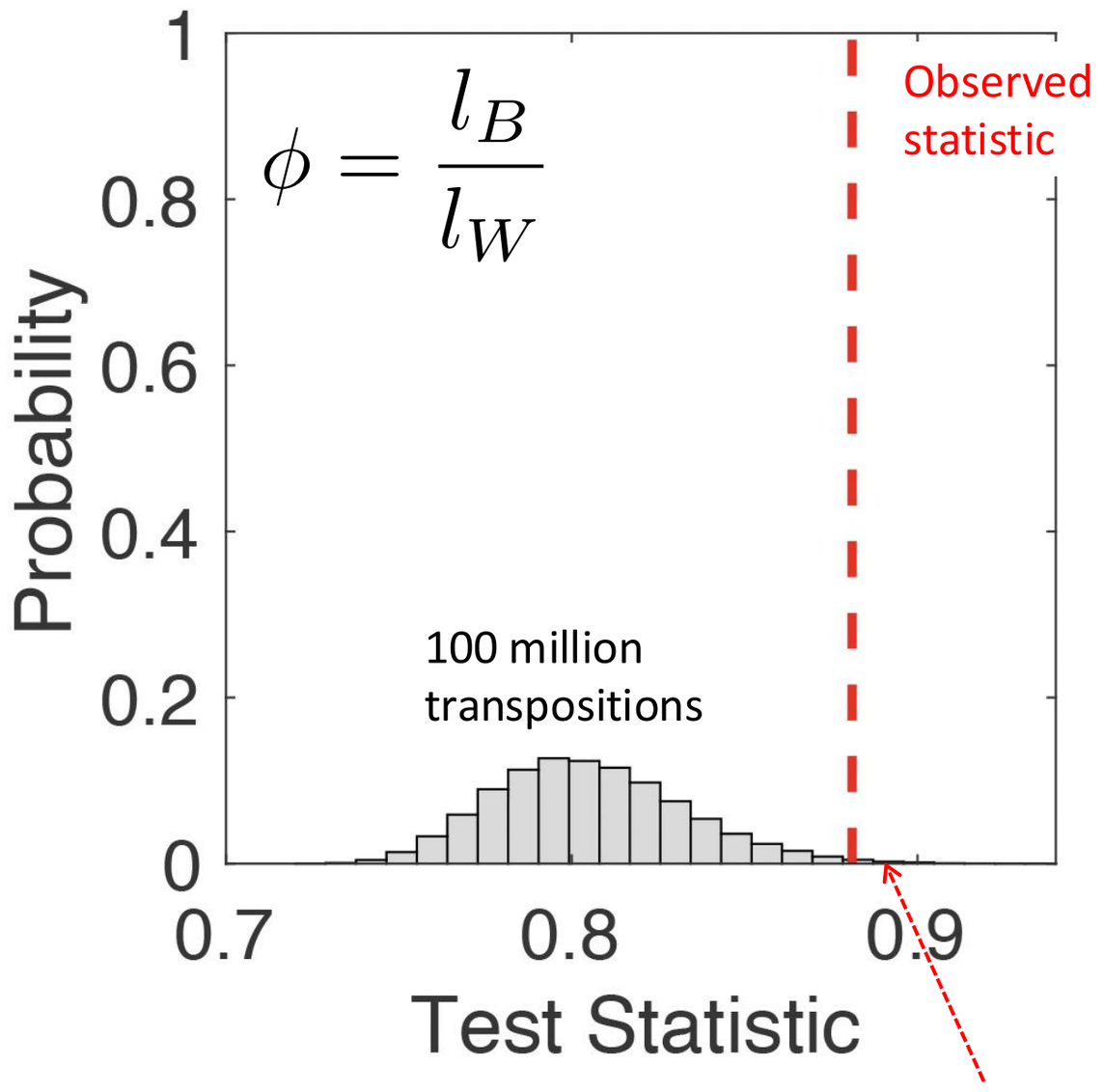


There exists a monotone decreasing function f satisfying p = f (A)

Sketch of Proof in Chung et al. 2023 NeuroImage

Statistical inference on the equivalence of clusters





P-value 0.0086

$$l_W \rightarrow l_W + \Delta(tranposition)$$

$$l_B \rightarrow l_B + \Delta(tranposition)$$

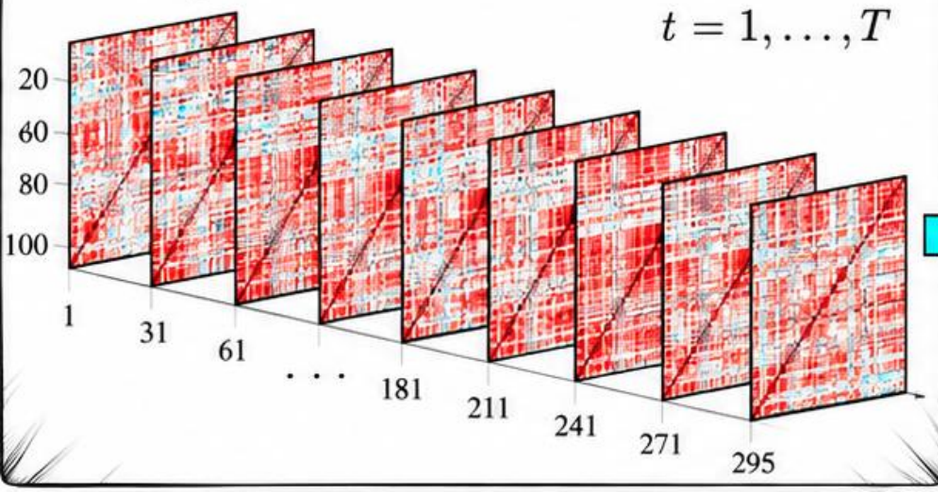
Songdechakraiwut and Chung 2023

k-means clustering to state space estimation

1. Time-varying connectivity matrices

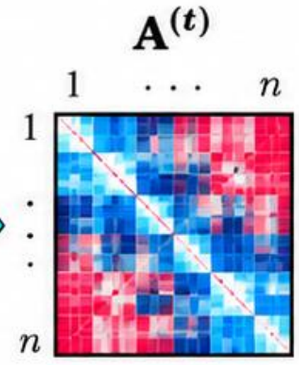
For each subject, compute a sequence of connectivity matrices over time.

$$\mathbf{A}^{(t)} \in \mathbb{R}^{n \times n}$$
$$t = 1, \ldots, T$$



2. Vectorize and normalize

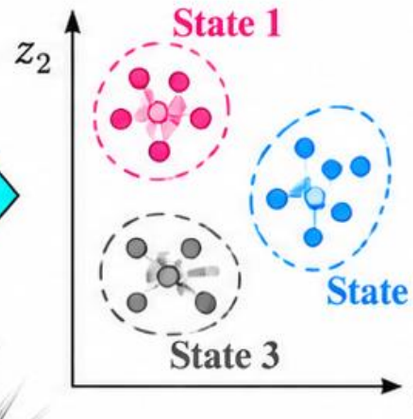
Convert each matrix to a feature vector (using the upper triangle) and normalize.



$$\mathbf{x}_t = \mathrm{vec}\left(A^{(t)}_{\text{upper}}\right) \in \mathbb{R}^d$$
$$d = n(n-1)/2$$
$$\mathbf{z}_t = \frac{\mathbf{x}_t - \boldsymbol{\mu}}{\|\mathbf{x}_t - \boldsymbol{\mu}\|}$$

3. *k*-means clustering in state space

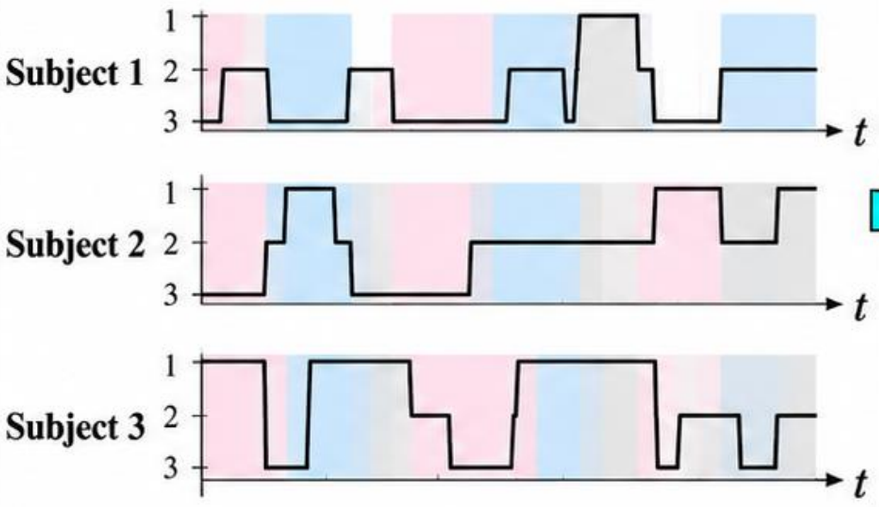
Cluster the normalized vectors $\{\mathbf{z}_t\}_{t=1}^{T}$ into K states using k-means.



$$\min_{C,\{\mu_k\}} \sum_{t=1}^{T} \|\mathbf{z}_t - \boldsymbol{\mu}_{c_t}\|^2$$
$$c_t \in \{1, \ldots, K\}$$
$$\mathbf{z}_t \approx \boldsymbol{\mu}_{c_t}$$

4. State sequence for each subject

Assign each time point to the nearest cluster centroid.

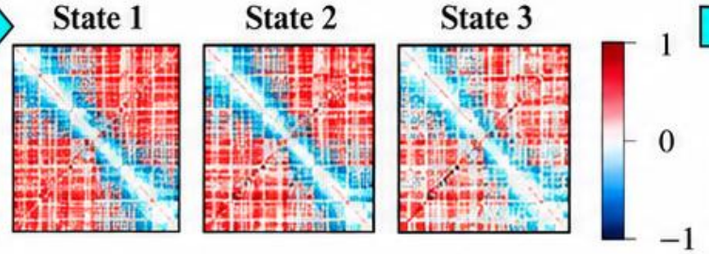


Each subject has a sequence of brain states over time.

5. State-specific connectivity

For each state k, compute the mean matrix of all time points assigned to that state.

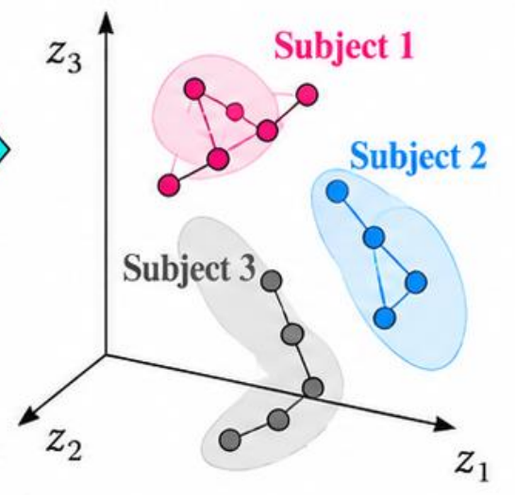
$$\bar{\mathbf{A}}^{(k)} = \frac{1}{|S_k|} \sum_{t \in S_k} \mathbf{A}^{(t)}$$



Each state represents a characteristic connectivity pattern.

6. State space and further analysis

Represent each subject by a trajectory in a low-dimensional state space and build a statistical model.

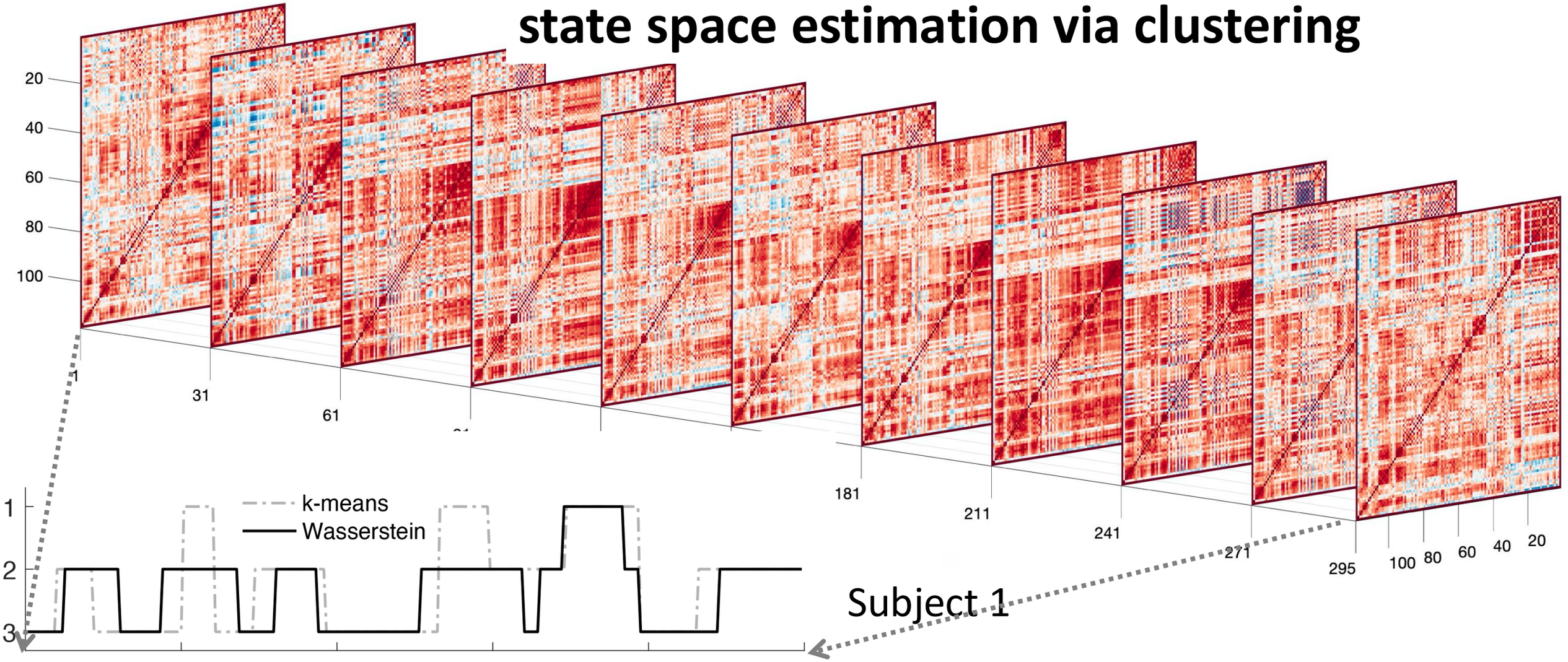


Applications:

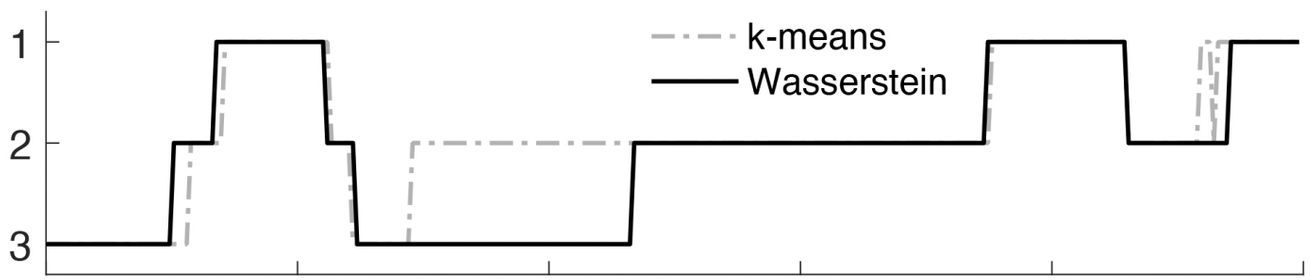
- compare state dynamics across subjects
- group-level modeling
- relate states to behavior or covariates

Problem: Is the state <u>reproducible</u> biomarker?

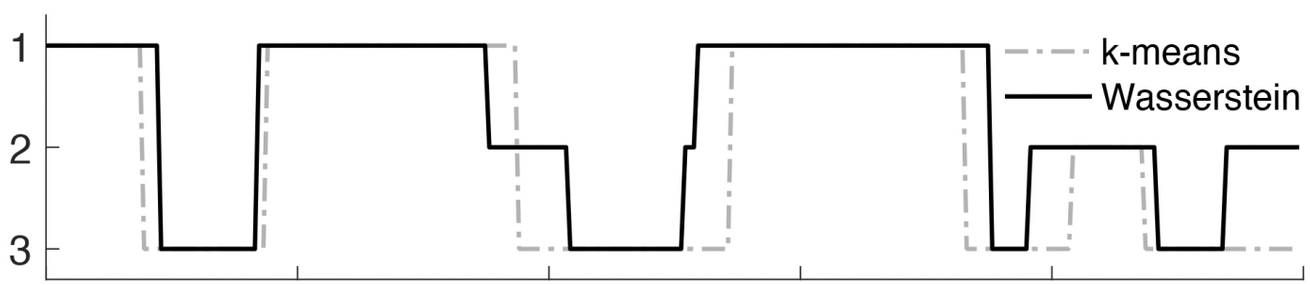
state space estimation via clustering



Subject 1



Subject 2



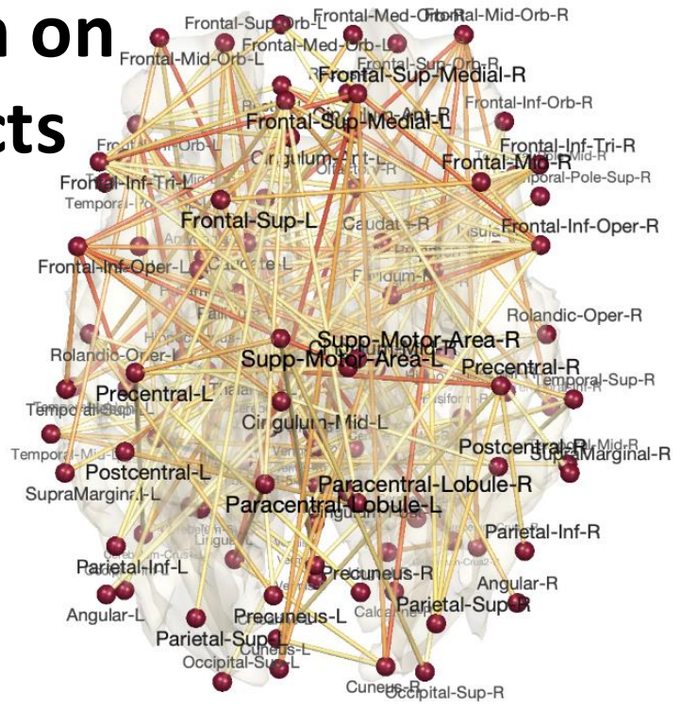
Subject 3

2nd level analysis:

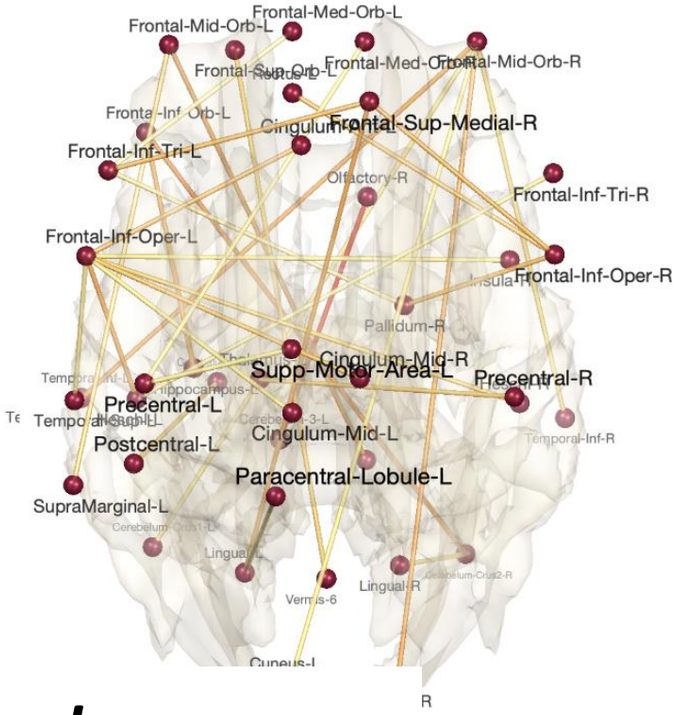
Statistical model

State space estimation on 479 subjects

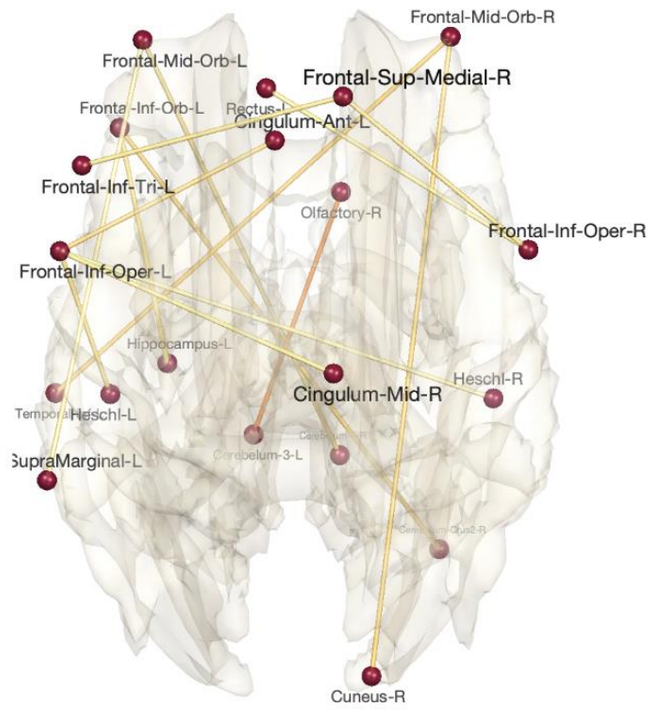
State 1



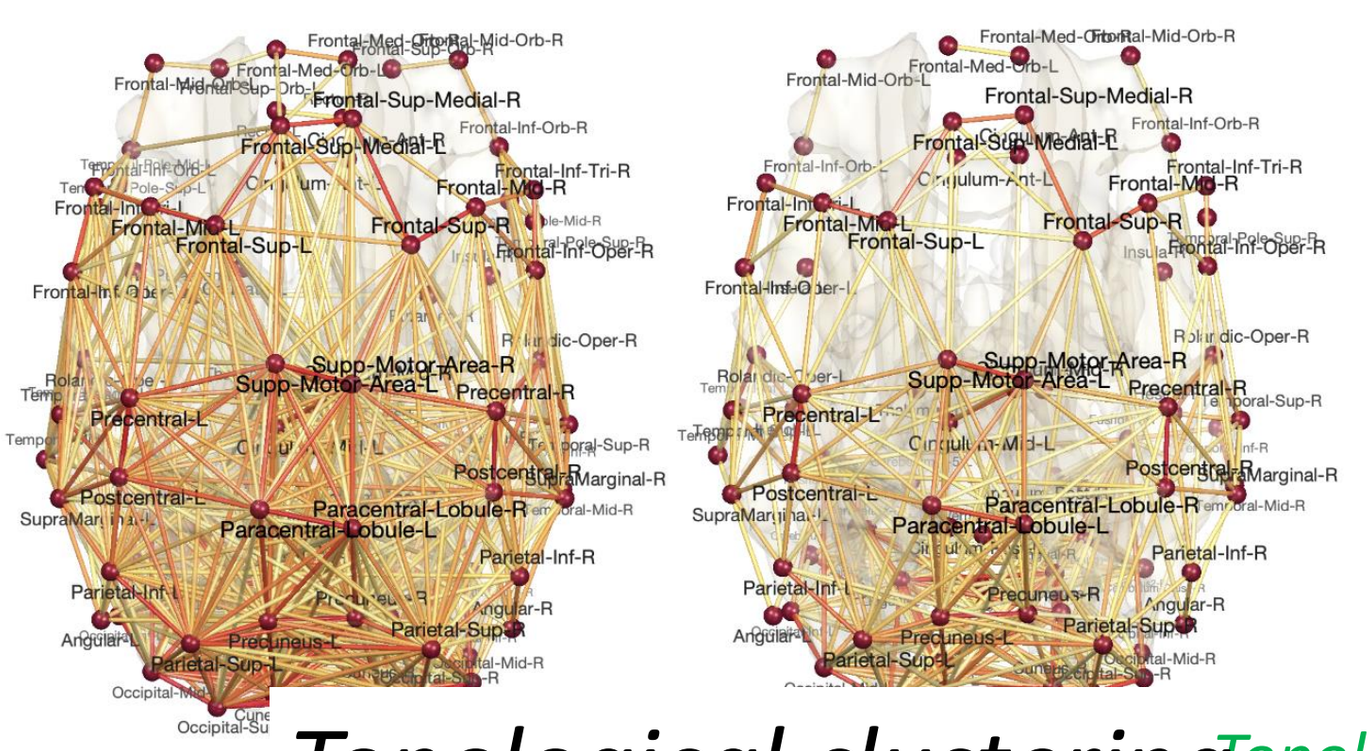
State 2

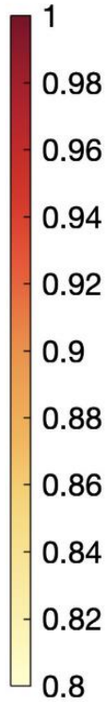


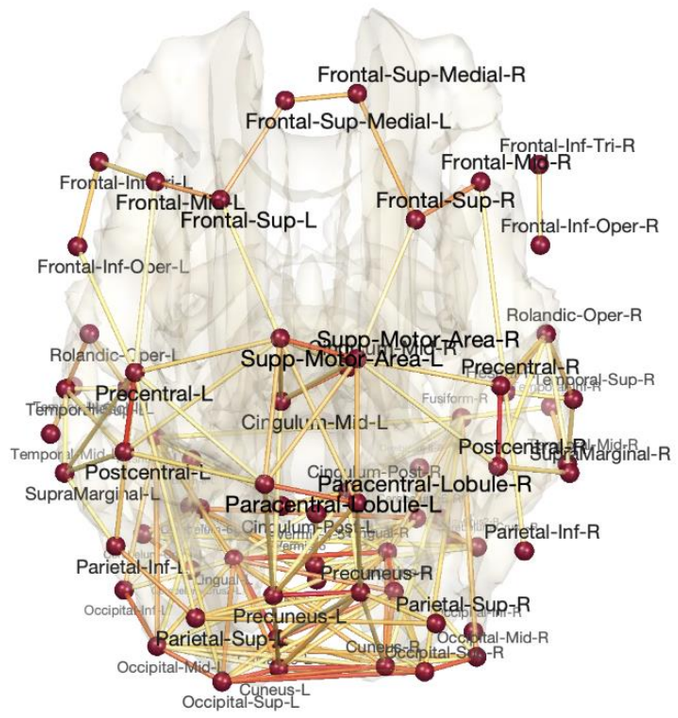
State 3



k-means *Sample mean in each state*



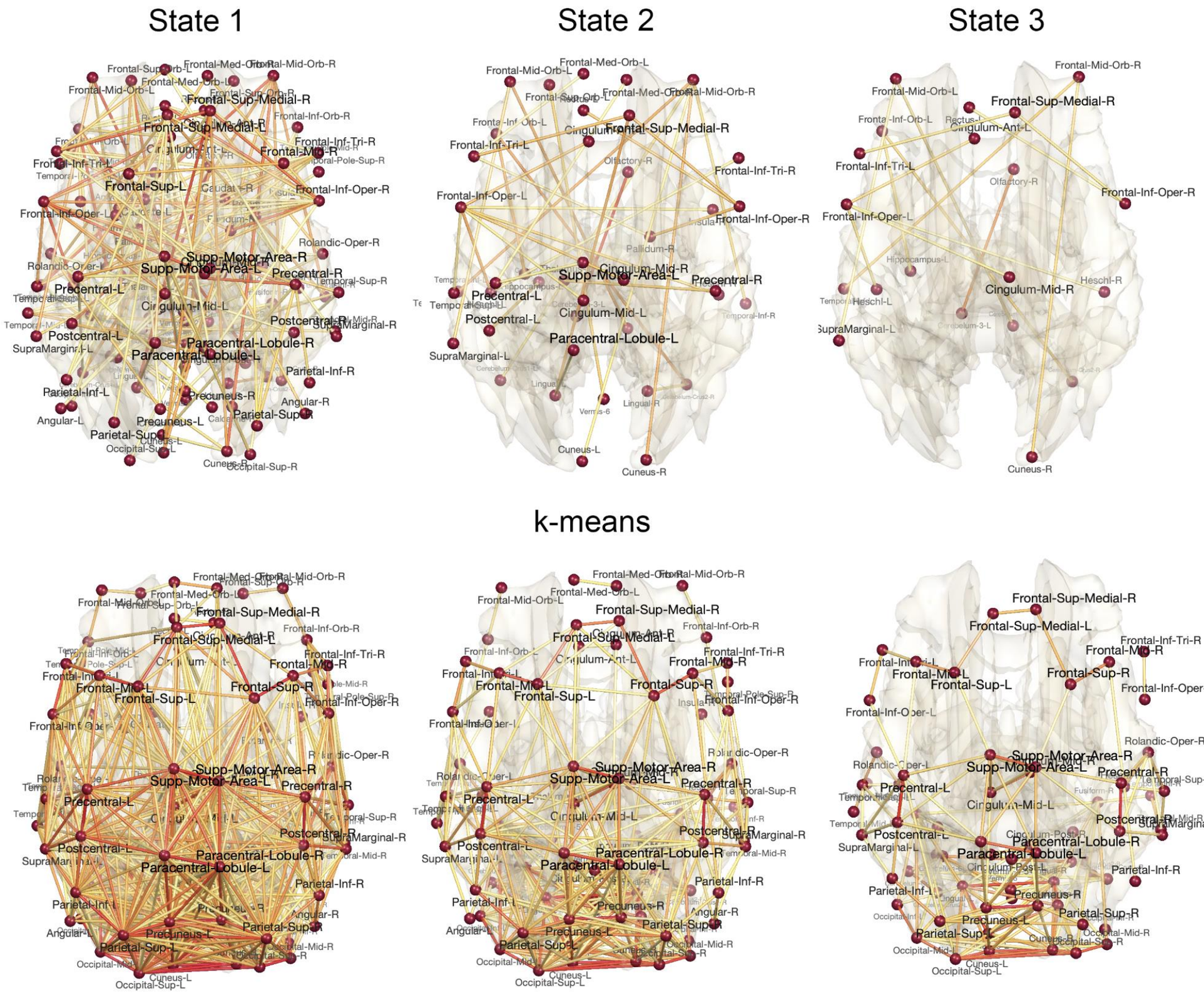




Topological clustering *Topological mean in each state*

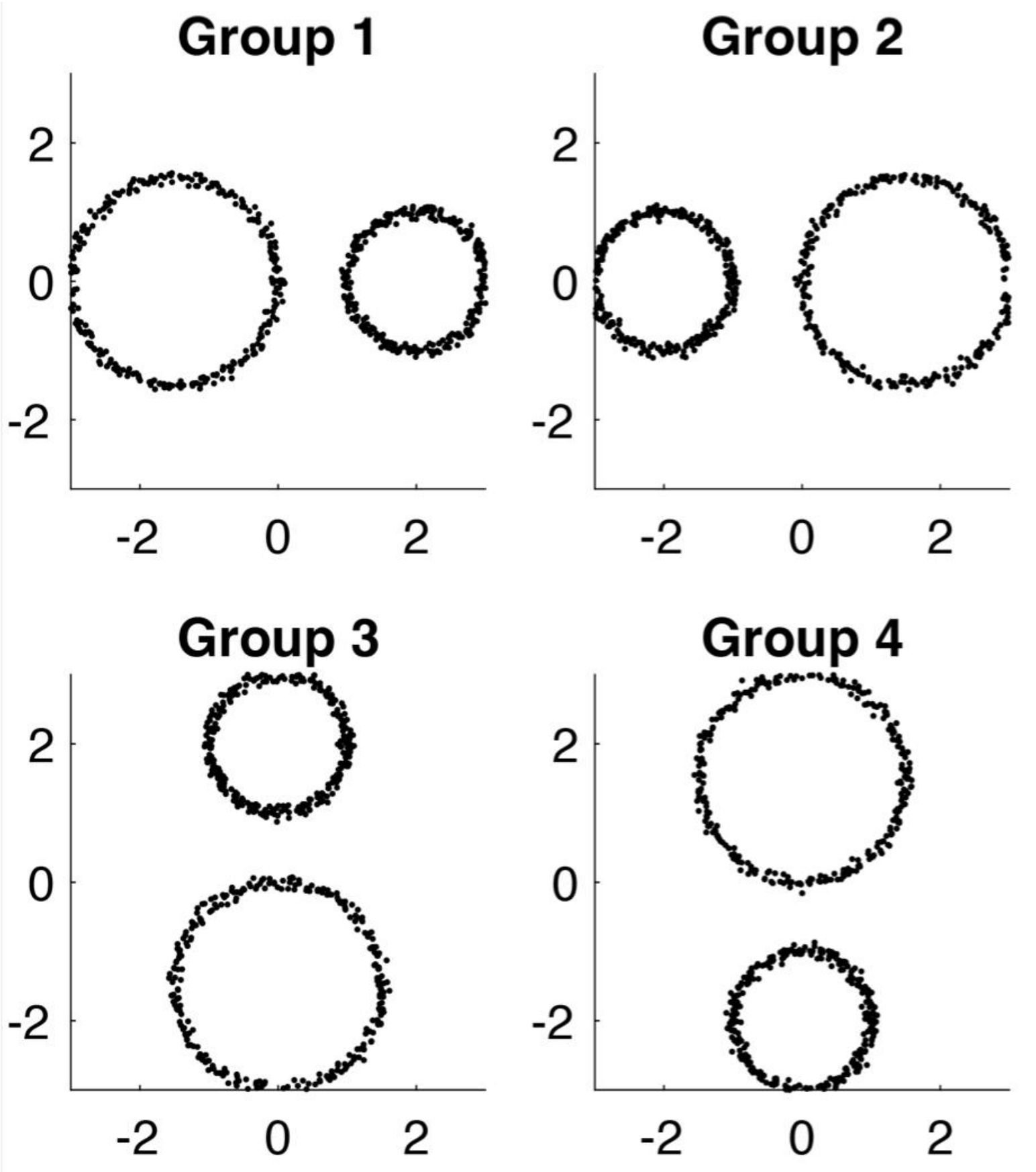
Chung at al. 2023 arXiv:2201.00087

But which method is more accurate or correct?



Often need <u>validation</u> on ground truth toy examples

Geometric clustering methods fail topological task



All false positives

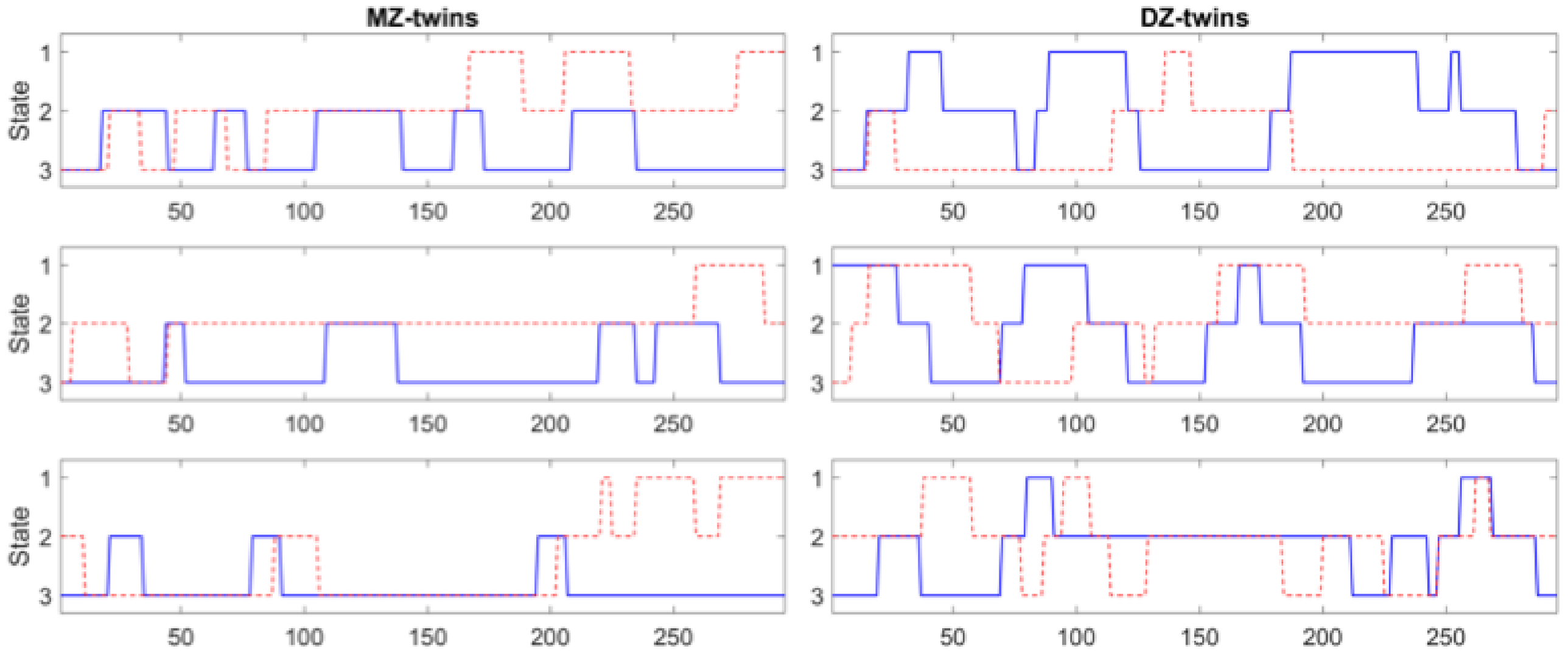
K-means clustering
0.98 +- 0.01

Hierarchical clustering
1.00 +- 0.00

Topological clustering
0.63 +- 0.04

MATLAB

Is the state-change heritable?



UW-Madison twin study (200 twin pairs)

Topological clustering can answer this question

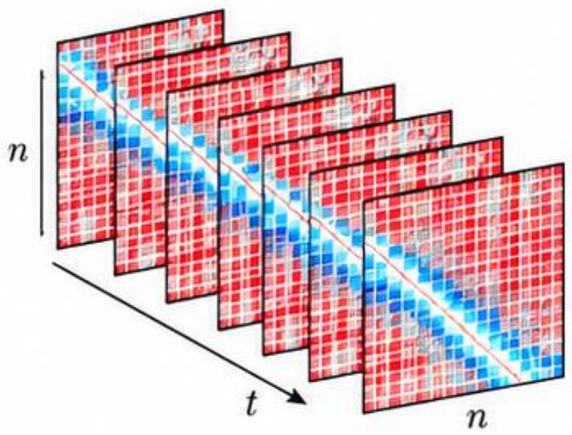
Chung et al. 2024 arXiv:2201.00087 (PLOS Computational

What is wrong with the state space estimation?

1 Time-varying connectivity

For each subject s, compute a sequence of connectivity matrices

$$\boldsymbol{A}_s(t) \in \mathbb{R}^{n\times n}, \quad t = 1, \ldots, T_s$$



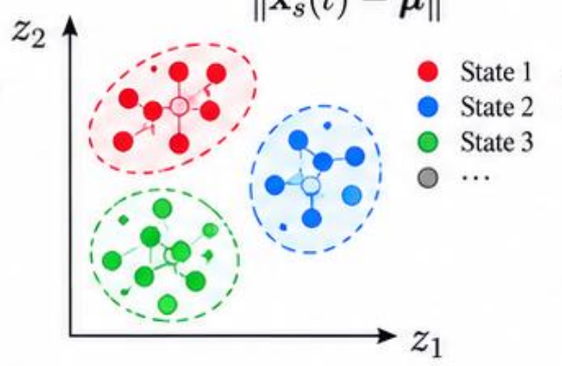
(e.g., correlation, coherence, etc.)

2 Clustering to obtain states

Vectorize and normalize each matrix

$$\mathbf{x}_s(t) = \mathrm{vec}\big(A_s(t)_{\mathrm{upper}}\big) \in \mathbb{R}^d$$

$$\boldsymbol{z}_s(t) = \frac{\mathbf{x}_s(t) - \boldsymbol{\mu}}{\|\mathbf{x}_s(t) - \boldsymbol{\mu}\|}$$



Apply k-means to $\{\boldsymbol{z}_s(t)\}$ from all subjects to obtain K cluster centroids $\{\boldsymbol{\mu}_k\}$.

3 State sequence for each subject

Assign each time point to the nearest centroid:

$$c_s(t) = \arg\min_k \|\boldsymbol{z}_s(t) - \boldsymbol{\mu}_k\|$$

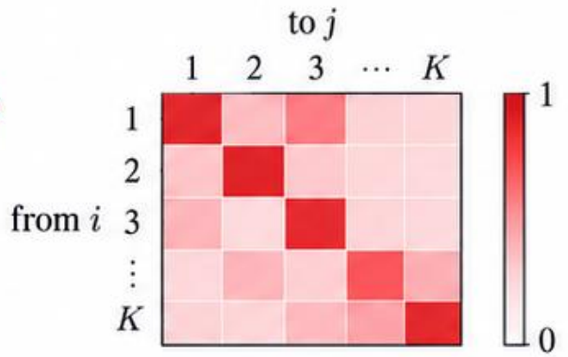


Each subject yields a discrete state sequence $\{c_s(t)\}_{t=1}^{T_s}$, $c_s(t) \in \{1, \ldots, K\}$.

4 Subject-level dynamics

Model the state sequence as a Markov process (or HMM):

$$P_{ij}^{(s)} = \Pr(c_s(t+1) = j \,|\, c_s(t) = i)$$



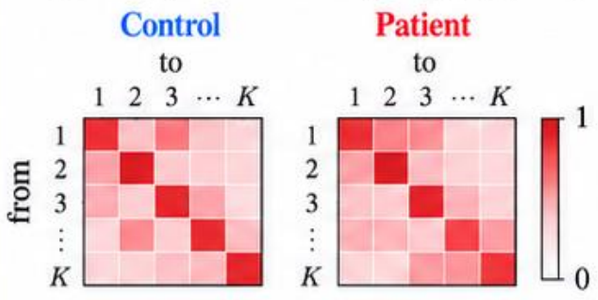
Estimate $\boldsymbol{P}^{(s)}$ from counts or using HMM to allow for noisy observations.

5 Group-level model across subjects

Model subject variability and covariate effects, e.g., multinomial regression:

$$P_{ij}^{(s)} = \frac{\exp(\boldsymbol{\beta}_{ij}^T \boldsymbol{y}_s)}{\sum_{l=1}^K \exp(\boldsymbol{\beta}_{il}^T \boldsymbol{y}_s)}$$

$\boldsymbol{y}_s$: subject covariates (group, age, etc.)



Test group differences in transitions and/or state occupancies.

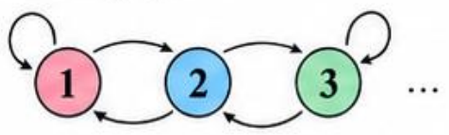
6 Additional state dynamics

Other useful quantities:

- **Occupancy rate:**

$$\pi_s(k) = \frac{1}{T_s} \sum_t \mathbf{1}(c_s(t) = k)$$

- **Dwell time:** distribution of consecutive time points in the same state
- **Transition graph:**



- **Continuous-time model** (optional):

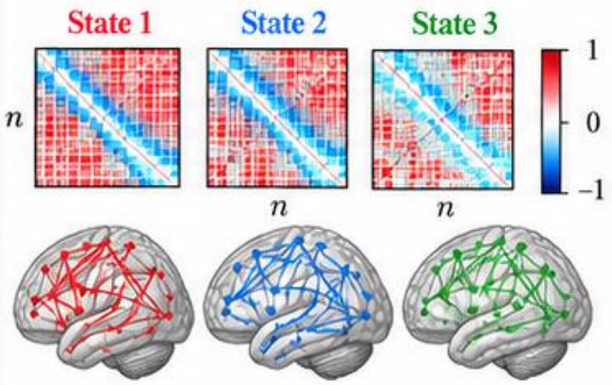
$$\frac{d\boldsymbol{p}_s(t)}{dt} = \boldsymbol{p}_s(t)\boldsymbol{Q}^{(s)}$$

(generator matrix $\boldsymbol{Q}^{(s)}$)

7 State-specific brain patterns

For each state k, average the matrices assigned to that state:

$$\bar{\boldsymbol{A}}^{(k)} = \frac{1}{\sum_s |T_{s,k}|} \sum_s \sum_{t \in T_{s,k}} \boldsymbol{A}_s(t)$$

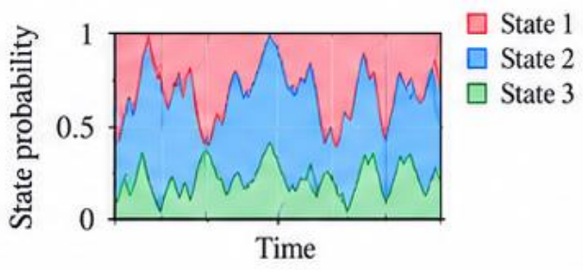


Each state corresponds to a characteristic brain connectivity pattern.

8 Analysis and interpretation

Use the learned state dynamics to:

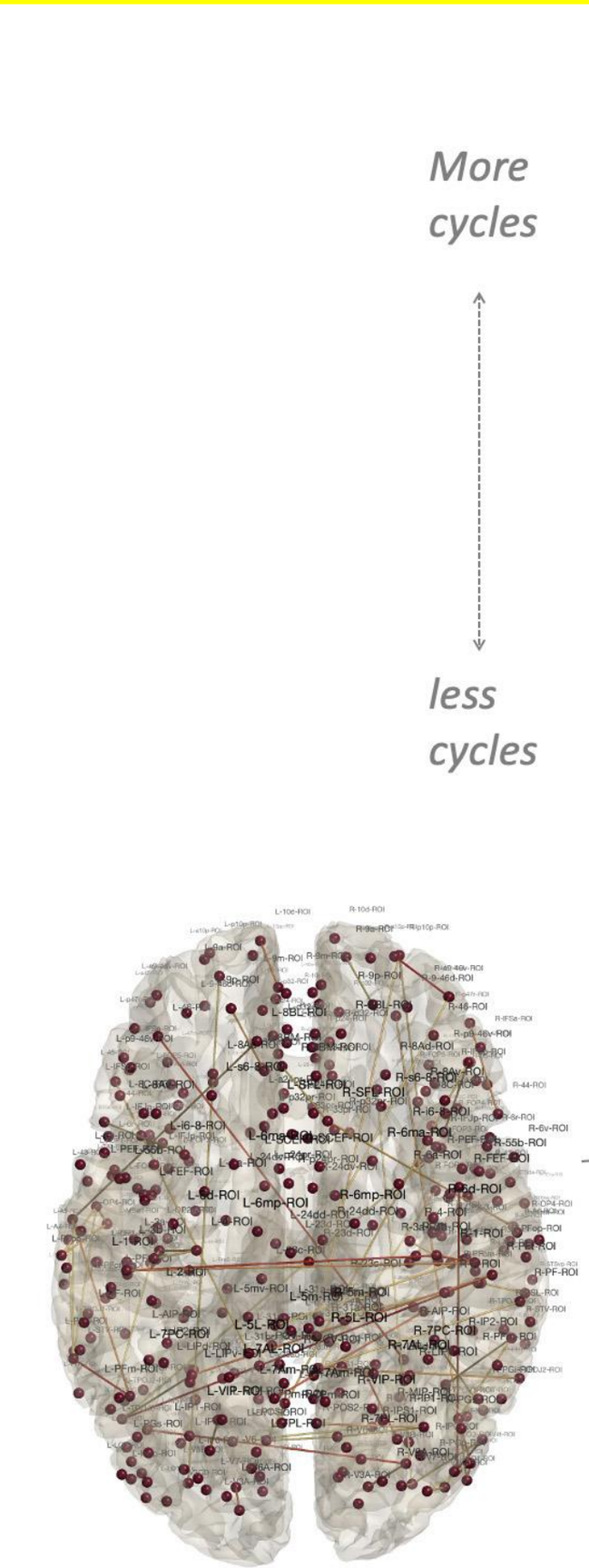
- Compare groups (e.g., control vs. patient)
- Relate transitions/occupancies to behavior or clinical variables
- Predict subject-level trajectories
- Identify biomarkers (e.g., state features, dynamics parameters)



Result: a principled, generative model of brain state dynamics across subjects.

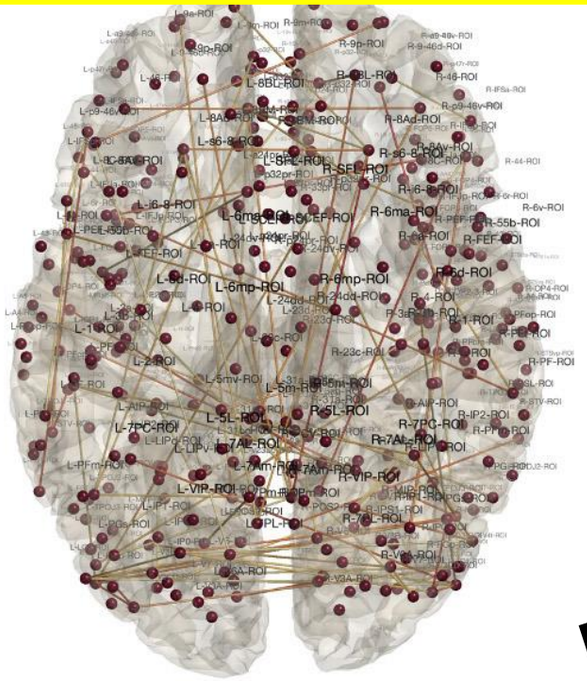
Project: Determine if the state change dynamics itself is reproducible.

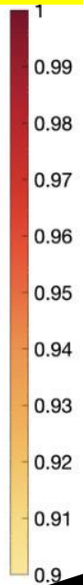
Interpreting clustering results is often needed

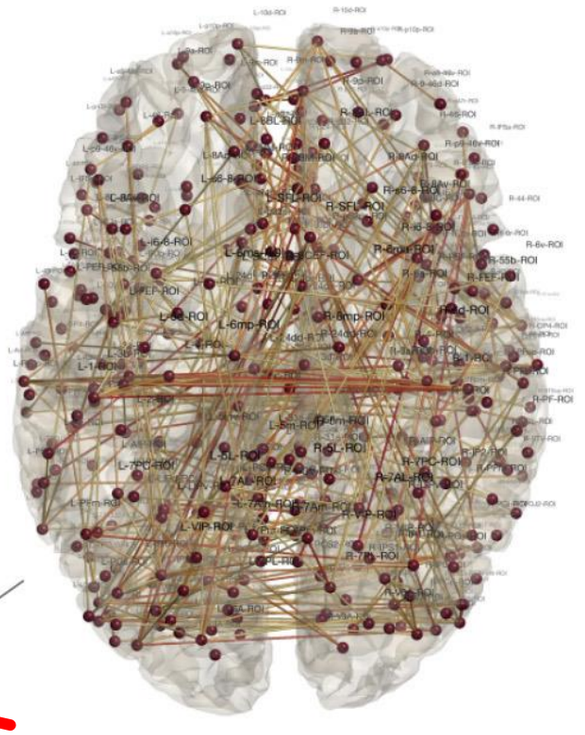


More cycles

less cycles





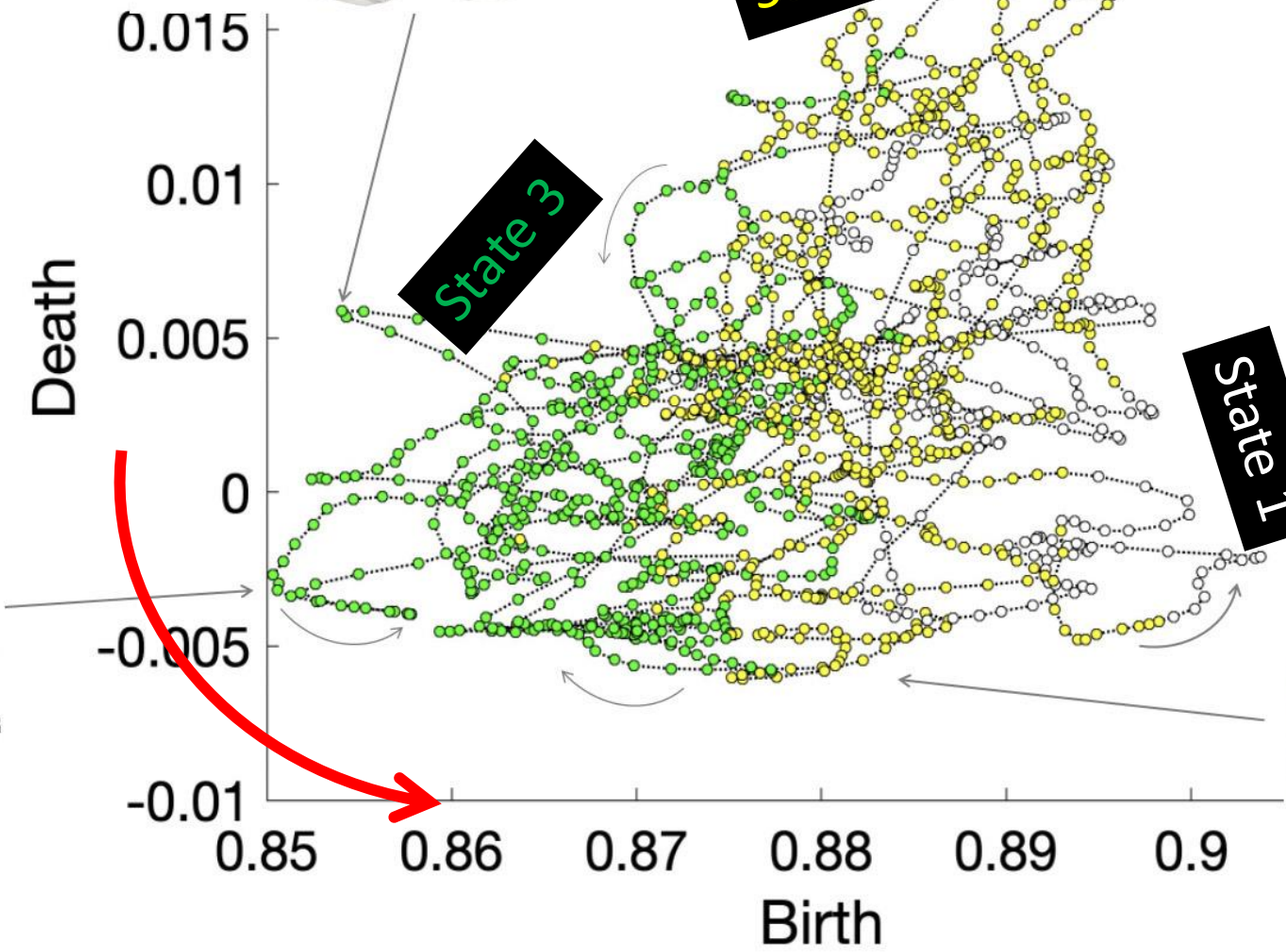


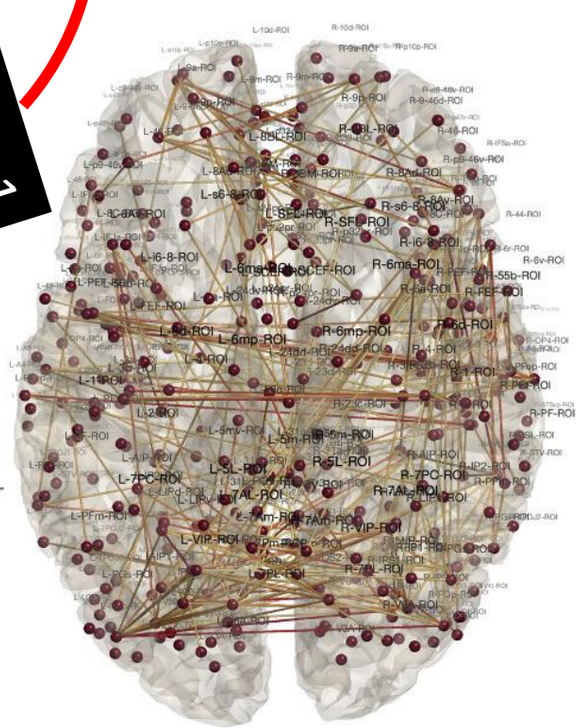
State 2

Mostly counterclockwise

State 3

State 1



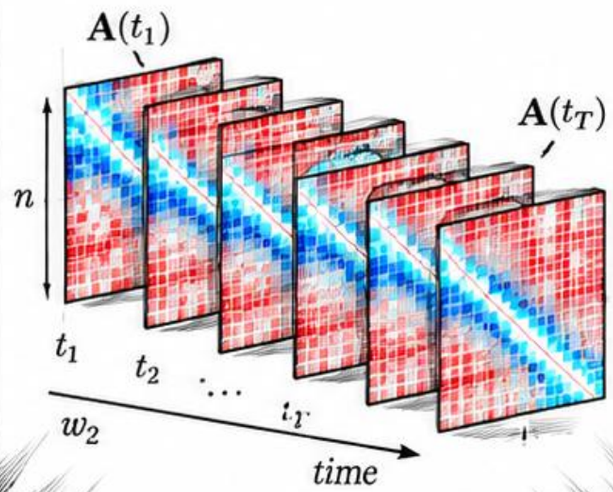


More disjoint subnetworks ⟷ *Less disjoint subnetworks*

Better approach: Dynamic state-space model

1. Time-varying connectivity

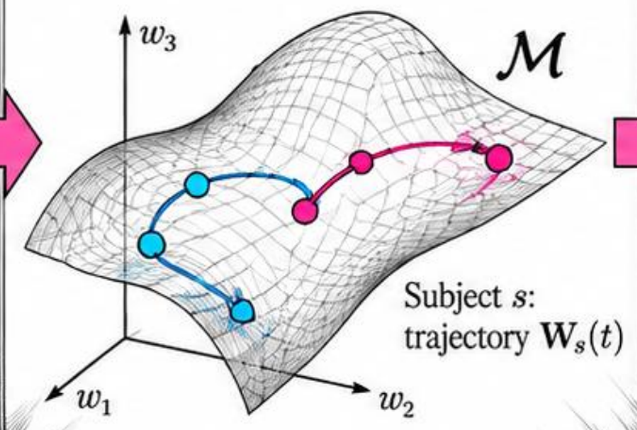
For each subject, estimate a time-varying connectivity matrix (e.g., correlation, coherence, or directed measure).



2. Continuous state space

Model the connectivity as a trajectory in a low-dimensional continuous space.

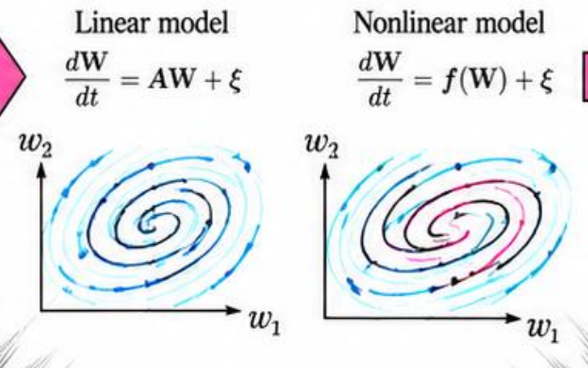
$$\boldsymbol{A}_s(t) \approx \boldsymbol{F}(\mathbf{W}_s(t)) + \boldsymbol{E}_s(t)$$



3. Dynamics model

Model the continuous dynamics
e.g., linear or nonlinear state-space model.

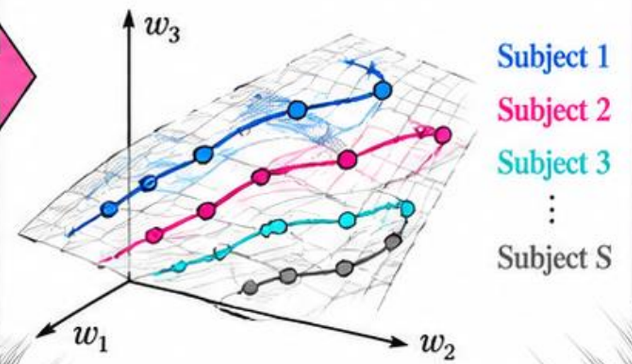
$$\frac{d\mathbf{W}(t)}{dt} = \underbrace{\boldsymbol{f}(\mathbf{W}(t);\theta)}_{\text{parametric or neural dynamics}} + \boldsymbol{\xi}(t)$$



4. Subject-specific estimation

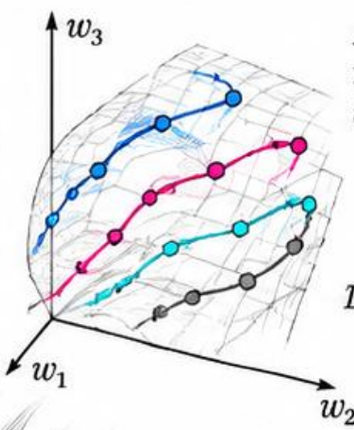
Estimate model parameters for each subject from the observed connectivity time series.

$$\theta_s = \arg\min_{\theta} \sum_{t=1}^{T} \|\boldsymbol{A}_s(t) - \boldsymbol{F}(\mathbf{W}(t;\theta))\|_F^2 + \lambda \boldsymbol{R}(\theta)$$



5. Reproducibility across subjects

Compare dynamics (not discrete states) across subjects using alignment and distance on trajectories or dynamics.



Are the trajectories reproducible across subjects?

Distance between dynamics

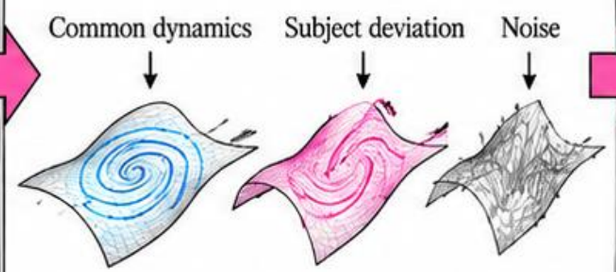
$$D^{(s,s')} = d(\boldsymbol{f}_s, \boldsymbol{f}_{s'})$$

(e.g., Wasserstein, Procrustes, or kernel)

6. Population model

Hierarchical model to capture common dynamics with subject variability.

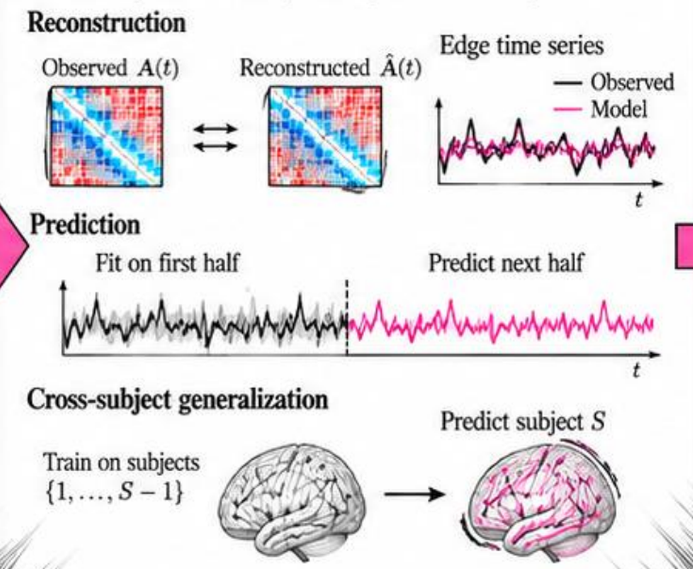
$$\mathbf{W}_s(t) = \mathbf{W}_0(t;\theta) + \boldsymbol{u}_s(t) + \boldsymbol{\xi}_s(t)$$



Share a common dynamical structure but allow individual differences.

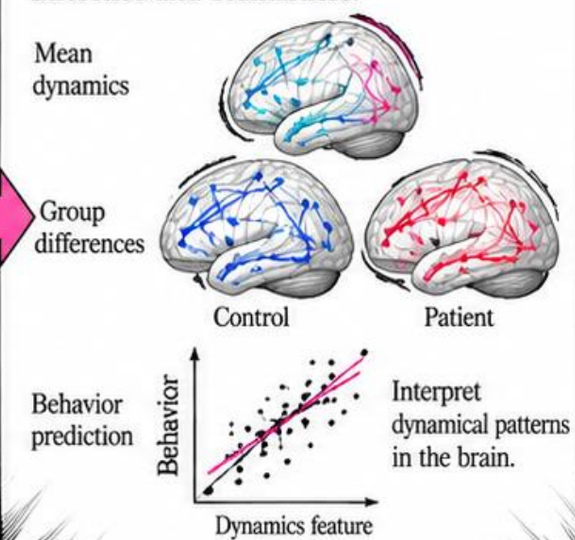
7. Model validation

Test reproducibility and predictability.



8. Applications

Use the learned dynamics for scientific inference and biomarkers.



Project: Determine if the state change dynamics itself is reproducible.

Learning outcome

Can formulate clustering problem

Can formulate topological clustering

Can understand the performance difference between topological vs. geometric clustering